Active Inference BookStream 002.02 ~ Parr, Pezzulo, Friston ~ Chapters 4, 5, 7, 8

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foreign 2023 and we're in active inference textbook group slash bookstream 2.02 thanks Ali for joining so what we're going to do today is give a short overview of the chapters from the par at all 2022 book we're going to do chapters four five seven and eight and we're just going to pause between them because then we'll clip them into the shorter videos append that to the playlist just so there's a first video overview of each of the chapters and this is the second in that work all right so we'll do chapter four um we'll just wait a few seconds and then start chapter four okay chapter four is called the generative models of active inference and it begins with a quotation everything should be made as simple as possible but not simpler by Albert Einstein Ollie what is your overview thought or warning for chapter four okay so uh after the preliminary materials um in chapters two and three uh which was basically largely based on um concept providing some conceptual framework for developing uh the further Theory chapter four delves into much more detail in terms of mathematical formulation and it unpacks a lot more uh the way that the central equations of active inference is derived and how to construct the important elements of active inference models uh so say matrices a b c and d and also how to put together generative models in different situations so it basically lays out the foundation or um for I mean constructing an active inference models uh both uh for discrete time situations and continuous time at once uh which will be used later in chapters seven and eight but this is uh probably one of the most uh challenging and uh at least mathematically dense chapters in the book so I would personally suggest reading through this chapter really slowly and even if we don't get to understand every single detail of the chapter uh obviously we can return required as we go through uh the textbook thank you Ali yes so let's look through the sections just to add on though chapter four is one of the larger and more equation dense chapters because it is the common kernel or basis that's then going to get applied in chapter five in the neurobiological case there's a recipe for making chapter 4 in chapter six that's the recipe for active inference modeling chapter 7 and 8 are about the discrete and the continuous time variant or or subtype or motif of these kinds of things called generative models so this is the real common root and we'll just look at what the sections are this chapter complements the preceding chapters conceptual treatment of active inference with a more formal treatment section 4.2 from Bayesian inference to free energy what would you say about this section Ellie okay so as we know the free energy principle is inspired by previous work on Bayesian inference I mean all the way way back to helmholtz theory about um unconscious inference or something to that effect I can remember the exact term but here I mean it provides in a more in a bit more detail how we can derive pre-energy principle uh formalism using the established Bayesian inference formulation and particularly one of the key uh movements or at least one of the key decisions uh in uh through the derivation of free energy principle formulation is using uh Jane's inequality principle to derive uh an upper upper bound instead of just using uh the exact values to compute um or or to achieve required parameters so uh that's basically uh in my opinion the key premise of uh section 4.2 and to see uh in a bit more detail how we can achieve those uh upper bounds using uh Jane's inequality uh directly from by using um I mean manipulations of Bayesian inference based in statistical formalism thanks I'll just add one point from this section broadly these are the problems of inferring states of the world perception and inferring a course of action planning so this is again referring to the perception and action and everything that happens in between is the internal or the cognitive part of the inference but this is like the blanket State cybernetic input output and then let's look at the first equation or how much equations overall or what equations do you think we should highlight uh okay so um maybe we can uh I mean uh just uh as a general comment about these um different equations uh well each of these equations provide a distinct step toward deriving uh the ultimate whole picture so uh even if we don't quite understand how we can derive from uh each step to the other one it's good to know that it's only required to understand how we get to that ultimate whole picture but ultimately what we would need in order to develop um for a develop active inference models is the ultimate equation or ultimate whole picture so this is just a way to elucidate uh the steps toward developing uh that whole picture but again it's not an essential requirement to understand um the materials of the rest of the book uh but if we go from uh I mean equations 4.1 toward the 4.4 or other words the pre A variation of free energy well equation 4.1 is just a basic definition of um probably some properties of probabilities in terms of conditional probability and so on so equation 4.2 provides the central chains inequality principle and how it relates to um I mean conditional probabilities and uh of course joint probabilities and then by using uh those uh two properties or those two equations we ultimately get to 4.4 which is the definition of variation free energy parameter uh which is the parameter of interest that needs to be optimized in order to inference to happen or at least perceptual inference to happen in active inference models thanks only thing all add is f is the letter used for variational free energy you can think of it like a computer program and the arguments that it takes in or the variables that it takes in are Q which is the distribution that's under the statisticians control and Y which are the data which are outside of the statisticians control do you want to describe more about anything in this equation or carry on uh just one thing to uh that can probably be helpful is uh to somehow compare these steps with um the initial picture we had from uh chapter two uh because uh variation free energy was first introduced in chapter two so uh it can be helpful to go back and forth between chapters two and four and try to connect the dots between um the related points there section 4.3 generative models all right I'll read the first sentence then you can give some thoughts to calculate the for the free energy we need three things data a family a variational distributions and a generative Model comprising A prior and a likelihood in this section we outline two very general sorts of generative model used for active inference and the form the free energy takes in relation to each but okay so uh as mentioned earlier this chapter deals both with discrete time and continuous time situations so clearly we would need two different types of generative models for each situation and obviously the the generative models or the way to construct generative models or discrete time situations uh would vary quite a bit from the one for continuous time situations um but the general principle underlying those General models are basically the same which is uh I mean to somehow construct a model of the environment and I mean either be it for the situation that that is sequential in time or for the situations that need to be somehow each moment of the situation needs to be accommodated in terms of a continuous time situation so uh figure 4.2 provides some examples of both so for uh yes let me see yes so we have some examples of different kinds of uh generative models some uh case case studies if you like and it provides various ways to show how the the dependencies between between variables can be modeled using these kinds of graphical probabilistic models so one common way to represent General to models is to use these kinds of graphical probabilistic models in active imprints literature which is at least in this case the circles would represent the random variables and the squares that would represent the distributions which would describe the dependencies between those random variables so we can see the clear relationships between those parameters here which is basically what this whole graph what constitutes the generative model that needs to be used for different situations and then in figure 4.3 we can compare the two different types of generative models based on uh whether it's discrete time or continuous time situations so the upper picture is a generative model for the discrete time situation and the lower picture is the parallel continuous time version of it and as we can see the general topology of these models are the same the only things that differ is the use of parameters for policies or I mean discrete time policies or the continuous time ones and we can obviously compare the different elements for for both uh priors States and uh external States internal States and so on by comparing these two models here yeah we often return to figure 4.3 it's kind of the Rosetta Stone of generative modeling for the context of this book because it's then going to develop out into chapter seven and eight and it represents a really fundamental decision made in modeling in the later chapters it's also shown how it can be made into a hierarchical model that combines aspects of both but within each level of modeling still these are the kinds of decisions that modelers are presented with when it comes to statistical modeling overall so section 4.4 goes into essentially the top half of figure 4.3 discrete time what would you say about discrete time okay so uh the discrete time situation uh is uh obviously the archetype discrete time situation which is upon the Palm DP models so at this point I would very much like to recommend uh following uh the steps following the material from step by step paper because uh that's uh in that paper uh the way to construct pump DP models is described in a bit more detail so if anyone feels like they should learn a bit more about the the gaps in the details uh I I would very much like to recommend that particular paper so uh yeah uh I don't know how much detail uh we should go into because uh it's I mean uh although uh it's not maybe a detailed enough for some tastes but it goes in a quite uh extensive detail uh about how we can construct these models using the the concepts we've learned uh in previous chapters uh so ultimately we reach equations for Point 13 and 4.14 which are basically the culmination of palm DP formulation using the vector notations and gradients and so on so uh that's when we go to continuous time situation great a few things intervene in the continuous time chapter ones that we'll just mention here because they're kind of boxed or partitioned from The Continuous time part about their following Pages first is Markov blankets we won't go into it here but kind of footnote that or look at some other places where we talk about it outside of this chapter overview figure 4.4 AZ message passing again a big topic let's kind of just go past it now we're back to the regularly scheduled continuous time generative model discussion and then another box to the generalized coordinates of motion so taking position plus derivatives of precision and that has some beneficial properties that are described and unpacked also elsewhere do you want to say anything about 4.5.2 uh well the only thing that comes to mind is although as I said before all the formulations here may look uh more I mean a bit too dense to understand at the first pass but uh some of the key uh maybe components here could be uh obviously the the material from box 4.2 and 4.3 uh is I think are quite essential to understand uh the uh the underlying principle behind uh deriving uh The Continuous time situation because without LaPlace approximation what we would have uh in terms of free energy minimization is uh it would look very much like the Gibbs it gives free energy so uh I mean the key distinction between uh the free energy Principle as described in active inference literature as opposed to Gibbs free energy is this distinction uh is the LaPlace approximation so this is what enables us to go from Gibbs free energy to um I mean uh the variation of free energy so uh yeah that's uh I mean quite essential to uh make this to be familiar with this essential approximation and obviously the concept of generalized coordinates of motion will come time and time again throughout the whole book particularly in chapters uh eight and nine uh so yeah those two concepts I believe needs a bit more attention so yeah sounds good box 4.3 LaPlace approximation equations another message passing representation and a summary the key message to take away is that approximate Bayesian inference may be framed as minimizing a quantity known as variational free energy this depends on the generative model that expresses our belief about how data are generated anything else you want to add ah nothing comes to mind at the moment because uh as I said this is I mean we're still in the stage that we want to develop our essential tools to be used uh in the rest of the book so uh here up until now I Believe by the end of chapter four we have acquired all the essential necessary mathematical tools and the next chapter chapter 5 is kind of acts like an interlude and I don't think it's the direct um I mean continuation of chapters uh one through four so I believe the the first um section or the first part of the book uh conceptually and mathematically ends here so yeah that's it yes it's a little bit like the pragmatic modeling part gets foreshadowed or explored in five now that we're all built up with four all right that's the end of the overview for four okay chapter five is called message passing and neurobiology what is your overview thought on chapter five okay uh I mean uh it's a kind of I don't know I have mixed feelings about this chapter because on one hand uh you see as far as I understand uh active inference uh although it originated uh as uh quote-unquote a unified theory of the brain uh I don't think uh it's a neurobiological theory per se uh of course there can be some correlations between a neurobiological uh components or concepts with active inference active inference um I mean Concepts but uh I mean it's not uh an essential premise of active inference Theory to provide a theory to provide a comprehensive theory about how the neurobiology of human brain or other organisms brain behave at it's a detailed and a neuro anatomical um level but at the uh then again it's nice to have these kinds of empirical correlations uh between the findings of neurobiology and the active inference Theory uh but I I don't think it's uh one of active um Central assertions at least uh to my understanding well said very interesting framing well chapter five definitely takes a very specific system of Interest approach by highlighting one of the most studied areas also one of the most relevant areas which is mammalian neuroscience and the chapter is going to introduce a few different motifs in the nervous system and essentially build up towards figure 5.5 which is at the end of the chapter and 5.5 wires together three specific neural systems that the chapter is going to focus on work in that area from so Ollie said it very well active inference was built up to in here is another level or type of science with assertions or with representations or mappings to any specific system but this is the kind of modeling that has been built up and done by fristin power pizullo and others over the decades with a focus coming from a human neuroimaging laboratory setting a lot of focus and study and attention and funding and everything on the mammalian nervous system but claims about the nervous system have have uh are not the basis of what active inference claims or how it's derived but this is like an example case study in neurobiology connecting back to some of the formalisms that we've just seen introduced in chapter four yeah and uh to add a minor Point uh to what you just said I think it's important to uh draw attention to the last sentence of the last paragraph of the first page it is important to draw a distinction between a principle I.E the minimization of free energy and a process theory about how this principle may be implemented in a certain kind of system so I I think uh this sentence here uh frames this chapter uh in relation to all the other technical chapters of this book so if every other chapter is about developing or at least up to now was about developing the principled formalism of active inference now chapter 5 provides a kind of preliminary sketch for the process theory of active inference which is uh obviously far from an extensive Theory it's just a single chapter but then again it can provide some important uh signpost signpost for anyone who wants to further investigate this area awesome free energy principle Bayesian mechanics all things in that area are on this principle not responsive to empirical data and then the process theory is about how the principle is implemented so the specific generative models that are made and how well they map or how well they do in a portfolio of models that can have very different goals and assumptions and all of this but the process Theory implementation lets us develop hypotheses that are answerable to empirical data like what is the kind of information or relationship between photons hitting the retina and changes in activity in neural systems and that's an informational question or can be abstracted in a way to an informational question that it turns out does have empirical support and results in unique explanations and predictions that doesn't mean that it always results but a lot of citations are provided here that's what we can explore in chapter the last paragraph of the first section describes that they're going to look at the three different neural systems okay section 5.2 micro circuits and messages what do you think Ali uh all right so uh I mean this chapter begins uh from how a message passing happens in neurobiological terms and compare it to the way active inference frames this message passing mechanism and specifically if we look at a figure 5.1 and compare this figure to the ones we've seen before in chapters uh in one through four I think it was in chapter four we can see some clear parallels between how this kind of cortical message passaging happens in the brain versus how it is framed in active inference literature and as we can see it's uh clearly inspired by the neurobiology of the brain uh but uh then it's important to keep in mind that it's not a direct one-to-one mapping between um these two models uh this is just um a kind of I don't know an interesting or Illuminating uh if you like parallel to keep in mind uh to somehow be a bit more confident about the viability of the theory we want to use for message passing and active inference uh which is to say it's not some haphazard Theory that's just been developed for practical reasons uh it has some basis in neurobiology although uh it's not necessarily fully congruent with every detail of neurobiological neurobiology great the specific example is going to involve this one region of mammalian cortex tissue that has these six layers and there's a ton of neurobiology the big takeaway for figure 5.1 is that it's possible to graphically lay out nodes and variables and find some empirical correspondences again some unique explanations and predictions in certain cases and that's one kind of modeling where it's really trying to understand uh and improve the ability to do correlation and intervention and counter factual causal type analysis with the real system of Interest or in a more pedagogical setting or in a research setting or an industrial setting you might sweep across large families of structures of models and there's no need to be grounded to any biological structure at all so this is just describing the specific neural anatomical research that really arose out of the Imaging work at UCL and SPM package that's where a lot of this comes from 5.2 yeah good and sorry just as a side note uh I think watching uh one of Thomas parr's lectures on neurobiology of active inference which is available in YouTube uh would really help to understand the materials of this uh chapter better so I highly recommend watching that one thanks figure 5.2 gives a re-rendering of a kind of classical View of a hierarchical predictive coding system works so here abstracting a layer from the tissue six layer to just two layers here computational layers now and then showing how there's hierarchical communication within a layer but also uh well there's they're signaling within a layer and there's a hierarchy uh in Bayesian modeling with variables that are higher order predictions about other variables and that's the basis of the predictive coding architecture so 5.2 looks at some ways that the something that resonates with the cerebral cortical architecture enables what might computationally look like or have some really strong and explanatory values in actually relating to computationally a hierarchical Bayesian model which could do various General tasks all right 5.3 is motor commands leaving the prefrontal cortex going down to the butterfly looking cross section here what is 5.3 okay so 5.3 moves to the other half of active inference framework which is uh I mean how how it can model uh the decision making and ultimately the movement of the agent uh in order to minimize the expected free energy as opposed to variation of free energy that we saw in perceptual uh half of active so it can provide a kind of uh correlation or analogy between the structural neural Anatomy uh particularly related to uh I mean the motor commands and uh how it can relate to active inference in particularly The Continuous time active inference so we can see that uh for I mean for the external uh uh event or I'm sorry for the external State we can take for example the proprioceptive afferent and then uh this proprioceptive afferent acts as a kind of why or the continuous time active inference which needs to be um I mean processed in in a way to optimize the expected free energy and how it relates to both attention and precision uh we will see a bit more detail about uh those uh those terms and the relation between them in chapter eight but I think here section 5.3 provides a good summary about the general uh paths through the motor command systems of neurobiology great let's say while the previous case study focused on how the connectivity within in between the cortical columns could have a computational relationship with a Bayesian hierarchical predictive coding architecture the argument of this second case study is that a continuous input continuous output kind of set point seeking reflexive motor Behavior with a moving set point with a descending moving set point enabling motion by changing ultimately the set point and enabling uh variation in the strategies to reach that set point through different mechanisms this is also describable in a compatible way that's a shorter section now section 5.4 subcortical structures okay so subcortical um structures are very important in the the decision making and and I mean the planning of the agents so uh obviously here we need to uh we need another kind of analogy between uh the that these plannings and decision makings happen uh neuroanatomically with the way uh that uh that it's framed in active inference but again uh we can see it's uh clearly based on uh I mean at least some of the important elements we've seen from uh the previous chapters uh so for example we we saw how policy is described or how how it relates to outcomes and and preference and so on uh we can see uh those elements are directly inspired by a neuroanatomical structures so I guess that's uh at least in my opinion this section here 5.4 uh seems a bit more uh sketchy in the meaning that it doesn't go into quite uh in the extensive details about how how how those structures can be um can be compared about for anyone who wants to uh further investigate these topics there are some useful references put on here on pages 93 and 94. so yeah thanks yeah it's really abbreviated and overviewed but we get an interlude from table 5.1 with putative roles of neurotransmitters so same perspective that we took before on neural anatomical functionalism here directly translates to neurotransmitter reductionism or essentialism or something like that so certainly all neurotransmitters and molecules they play variable roles in different settings this is the knee in scruffy manifold all over again one person might say well we we need a theory for every acetylcholine molecule in the world they're all in a unique context and someone else says all neurotransmitters are described by one parameter in this model I'm getting value from it so to me that's an account and somewhere in between is the work in this space which is making an attempt to have a principled and falsifiable approach to model the computational aspects of specific regions and contexts and settings and so acetylcholine noradrenaline dopamine and serotonin are given a little mini review here and so it's not an exhaustive or an exclusive claim it's kind of a provocation from computational and molecular neuroscience and people can look into the papers and also ones that probably have been published since 5.6 goes to continuous and discrete hierarchies which is graphically overviewed in figure 5.5 so what would you say about this uh yeah one interesting thing about this section is the observation that our lower level engagement with the environment uh it can be most successfully characterized with continuous time formulations but as we go up uh on uh I mean on the level of uh cognitive Concepts or at the level of cognitive hierarchies and we come to Concepts such as I don't know decisions or even beliefs and so on we can reach the area that the discrete time situations uh would probably be more efficient to to characterize the behavior of the agent so this multi-scale [Music] multi-scale structure of active inference modeling is quite evident in the way that our message passing happens in our brain in terms of our lower level data processing uh up into consolidating the higher level cognitive Concepts and ontologies awesome thank you to me figure 5.5 demonstrates the kind of whole of body approach that you could imagine there's so many organs and systems and phenomena for which there aren't specific generative models so little can be said about situations where no generative model has been articulated and here's one where it has so it gives you also it's kind of like reading a drosophila melanogaster review paper relatively it's like this is how much work it takes to get to this state of knowledge in an insect so then in another insect do we know less about that insect empirically and genetically so consider this to be what's known to be a lot however also about one of the most sophisticated or specific cognitive systems at least we know so there's that additional kind of like self-reflexive aspect to this chapter that is not a Cornerstone of active inference but here it's just presented in a synthetic case study anything else you want to say about five uh nothing particular comes to mind all right okay chapter 7 is called active inference and discrete time chapter seven is the first in a pair of chapters with chapter eight on discrete and continuous time so that they're kind of like two forks of a river that we discussed in chapter four and before and describe the recipe in chapter six now seven and eight are kind of like one level deeper going from the kind of all of uh this group of animals to one level deeper into its classification scheme on the way to the specific generative model for which it's actually given in its totality but everything prior to that is about the learning about its principles and this is kind of on the trunk of the path to discrete time modeling just like chapter eight will be about continuous time modeling what would you add in okay so I think uh chapters seven and eight uh really helps to understand um in a more practical way uh how the materials from uh particularly chapters one through five applies in real time situations uh so uh even if uh we somehow didn't get to understand every details of chapters one through four uh by when we come to Chapters seven and eight uh I I think uh I mean some of those uncertainties uh about or about our understandings uh can be uh clarified uh in a in a at least in a practical sense so uh I believe these two chapters are really helpful um in order to consolidate our understandings from the previous chapters awesome well said so it's going to involve specifying some discrete time models 7.2 goes into perceptual processing and the general structure of the chapter is going to walk through a series of examples that building complexity where they first start with perception in 7.2 introduce decision making and then describe a few more types of motifs or cognitive structure or patterns and also check out step by step and model stream one where it's built up to in a different way so the first example is I'll let you describe it since it's Musical okay so yeah the first example is um the situation in which we try to describe the performance of an amateur a musician in terms of how how we listen to the performance of an amateur musicians in terms of the the predictions we get from our anticipation of the following notes as opposed to the actual notes that's being played so these kinds of anticipatory reaction uh listening reaction to the musician can be successfully formalized using um discrete time active inference by uh putting up uh by by putting together the matrices a uh for the states and uh Matrix B for the transition between the states um or the transition probabilities which uh describe in this case describes uh the probability from going from one note to the other and uh obviously the um the actual sequence that's been played which can be described with uh The Matrix T so uh and another Point uh I wanted to point I wanted to mention is for anyone who has downloaded this chapter uh before I don't know I think about June or something I recommend re-downloading it from mit's website because uh they have corrected some of the typos that was previously present in this chapter particularly in gear 7.2 cool so this graphical model where a person is listening this is a general perceptual Bayesian framing it's specified just like with any other equations there's a lot to look into but a indicates the probability of an outcome given a state this is saying if it were all on the diagonal like an identity Matrix this is kind of a common Motif then States kind of map to themself so in the context of uh in the context of this model A represents the mapping between the observed note and the underlying hidden true note FNB describes the transition Matrix of how those change to time D is the prior they're specified figure 7.2 you want to describe it so in figure 7.2 or at least the incomplete version of uh figure 7.2 we see here uh well at the upper left uh part of the picture uh we see uh I mean uh the beliefs about uh about each note uh at each step at each time step and at up upper right uh we somehow translate those uh beliefs into a specific numerical values uh so uh so instead of just assigning some continuous values we uh we we've simplified the situation by assigning some discrete numerical values for each note and then the lower left is supposed to show the uh the act the free energy gradients and over time uh or in in other terms uh the prediction errors uh we get from um I mean comparing our predictions with the actual outcomes uh so lastly uh lower right picture uh shows uh in parallel to the upper right picture uh determines uh the values of these errors so uh we we can see both uh the The Continuous the initial or at least initial uh continuous assignments and values and then the further uh discretizing of the values in order to get the discrete time situation or the more tractable at this group time situations okay so it's a general passive inference task where there's priors about how states are going to change through time and then there's real data coming in so that's the kind of classical predictive coding video compression common filter Bayesian setting 7.3 introduces a key Motif which is decision making and planning as inference so this is the idea of having a base graph where the variables can relate to different things there's high composability and here the idea is that a variable is going to be proposed that we can do inference about that describes the process of decision making or policy selection so what would you say about 7.3 okay so 7.3 is obviously similar to what we saw in chapter 4 and if I'm not mistaken even the topology is exactly the same with that picture we saw previously so uh yeah this is the initial setup to or which acts also as a review about how these different components of General upon DP General to models needs to be described in in such situations but uh ultimately the specific case study we come across in this section is uh the attempt to model uh the behavior of uh of a of the mouse in a teammate so rat and a teammate so um especially uh teammates containing uh an aversive stimulus in one arm and an attractive stimulus on the other so this is this can act as a kind of toy example to use uh this kind of probabilistic modeling to uh describe these situations thanks so that leads us right to figure 7.4 here's a visualization of the situation with the the rat in this case where there's a a pleasant and aversive stimuli on each end of a decision point and there's also a epistemic opportunity to receive some information about the context that the animal is in and so that setting is described for both the case with white on the left block on the right and block on the left white on the right and those are shown in terms of their differences in the matrices the explicit specification of the generative model visualizations show some of the slices of the B variable which reflect different transition probabilities C represents the preferences which are expressed over the observable States D reflects the priors on the different states that need priors 7.4 what would you say about this yeah okay so in 7.4 it builds up on the previous section and adds other elements that we previously saw in chapters three and sorry two and four uh which is how the exact formulation for expected free energy can be used sorry variation free energy can be used to formulate the trade-off between uh the uh I mean information uh seeking and and or at least uh between the epistemic value and information seeking so uh here it uses uh again that rad example in a bit more uh more extended and elaborate form to formulate the epistemic value of observing a q in a given a given location and figure 7.7 is a representation of this situation but another situation that's been uh let me see yeah in 7.9 another case study discussed here is uh is is the situation of uh The Psychotic eye movements and um because it is something that can be quite successful you just describe or characterize in terms of information seeking versus the epistemic value and the situation here is uh let me see yeah shown visually in figure 7.9 which clearly shows how our visual uh psychotic eye movement eye movements can be described in such a way as to as to kind of trace the trajectory uh of our eye movement among different regions of the visual space so uh and how the information We Gather from a given region can be can affect the um I mean the subsequent trajectories of our psychotic eye movements so yeah that's basically the main premise of this section I guess nice great 7.5 what would you say about it okay so uh 7.5 again adds another dimension uh to the previous formulations and this time we get to update the generator models by learning and the so the the generative models for this situation is a bit more complicated than the previous ones because it now needs to account for a mechanism or a way to update the matrices we had before so in the previous situations uh we we didn't account for learning per se but here we directly update our general sorry the word update can be confusing here we we get to somehow improve our generative models to accommodate for these updating accounts and yeah so uh this the situation here uh or the case of study uh uh here uh which uh somehow elucidate the way that the learning um can be a common can be accounted for with these models is uh again a toy example of a creature uh in a in a simple world of uh black and white tiles which kind of tries to find a a path to reach a given destination a certain destination so it is more complicated than the situation we had for the rat example because it only had um I mean simple trajectories that needed to a Traverse but here uh the creature or the agent in this case needs to do lots of a lot more learning and information seeking and so on so all the previous elements uh is kind of combined in this example and it's a really good example to see how the different components of active imprints can be connected to each other nice and seven six hierarchical or deep first a box 7.3 interlude on structure learning boxed off topic and a lot to say but structure learning broadly refers to learning the structure about a model using the same types of methods that you might to do inference on for example a more observable sensor data reading something like that this section Works towards the idea of nested inference or or multi-scale modeling what would you say about figure 712 so again this situation is um I think the most complex situations of this chapter which builds up uh from the previous sections uh and this time it adds another layer to accommodate for for the inferences that happen uh in uh I mean different time steps so in this case we have multi-time or multi-scale inference and learning happening um both at the levels of um sorry at the levels of learning and at the levels of information seeking so this is represented in figure 7.12 which represents how uh kind of this fractal generative model can be seen as a component in um this multi-scale bigger generative or and as a kind of leaf in this bigger bigger generative model uh so it can be seen as a lower uh level inference happening at the leaf level going up to the hierarchy and influencing sorry uh collaborating on the whole process of learning and inference at the higher level so yeah yeah I guess that's somehow summarizes this figure so if you have anything to add that's that's great it's an example of the composability of generative models what we've uh talked about and had Toby Sinclair Smith describe as as the compositional cognitive cartography and just what kinds of connectors can and can't you do and how can that Motif that the discrete time model introduces and then the rest of these features including action and learning and so on get layered in on top what can you do with that 713 gives another example do you want to say anything about it or maybe continue on uh yeah so the case study here is the example of um linguistic I mean language learning uh through reading so not language learning uh maybe it's just what happens uh sometimes yeah yeah in comprehension so what happens when reading and in a anticipatory way uh the the words that that comes each after the other so uh why this kind of situation uh can be most successfully characterized with this kind of modeling because it involves a different scales of learning and comprehension both at the level of I mean reading at the level of uh somehow observing the letters and then going onto uh the words and then word groups and so on uh so uh yeah that's uh really interesting way to uh again combine all of those elements uh into a single unified model to see how those different time scales slow and fast time scales operate together to to build this more encompassing model of more incoming generative model of the pray any closing thoughts on seven uh nothing particular now thanks next chapters chapter eight which is going to go into the continuous time all right chapter 8 is called active inference and continuous time begins with that Timeless quote everything flows nothing stands still what would you say about chapter eight all right so this chapter uh probably is at my most favorite chapter in the book uh because of my own personal interests in um uh I don't know the process materialism and so on but uh yeah so chapter seven and acts as a really good starting point for anyone who wants to develop the discrete time situations and to model the screen time situations uh enacted within active inference framework but in chapter 8 we kind of get into get to a Model A bit more interesting or let's say more involving situations uh and they're not necessarily a kind of toy examples we saw at least at the beginning of chapter seven obviously as the title suggests this chapter deals with the continuous time situation so uh in the in that case we'll need to and maybe at this point refresh our memory about what continuous time situation involves by reading the relevant Parts reading Arabian relevant parts of chapter four so uh yeah in chapter four we saw that the generative model for continuous time situation uh derives derives from uh the uh it was a stochastic calculus in terms of uh putting the whole process um into two elements uh two stochastic um two elements of stochastic equations uh one of which is the actual uh State and the condition of actual States um or the behavior of the actual States and the other one uh is the randomness that we need to account for in each real-time continuous time situations so that's what we get here in equation eight a 8.1 and then building up from that equation uh we um it generalizes that equation to involve uh I mean the fun the functionals of G and F instead of just um the the single valued functions of G and F so uh then we get to put that into uh the situation that can be used for describing the behavior of dynamical systems which is a very well known uh situation to use these kinds of uh stochastic equations and it's widely studied how those those kinds of Dynamics can be characterized especially in recent Bayesian mechanics paper by Dalton exactly vadavel and others so and then it gets to some more specific examples such as a lockable terror Dynamics and the synchronicity and so on in order to show how these kinds of Dynamics can be elaborated upon and can be generalized to and and enables them to characterize a more complex situations so get that's a really short and brief overview of the whole uh chapter maybe uh we can talk about a bit more details as we go through right well said well I'm sure for another day the philosophical implications of a seven and eight and high road and low road and all these other parts of the textbook great topics um I I agree I would see chapter 8 as demonstrating continuity with some classical continuous time modeling motifs from a few different areas of dynamical Systems Science which is applied in like many many many fields but these are some classical examples so figure 8.1 goes a little bit more into depth or at least into more formalism detail about exactly what we saw in chapter five with the spinal reflex arc with the proprioceptive data coming in and then a differential being calculated with the set point which reflects a descending prediction from a decision-making layer and that can be viewed as this kind of mechanics that plays out in a phase space in continuous time like a spring moving around with with someone making a certain path with an attractor and a spring being dragged around something in that area box 8.1 goes into a very fascinating topic do you want to describe it uh well uh it's uh maybe uh one of the most um thought-provoking uh section pages of the whole book and if I remember correctly uh in all of the cohorts this particular box I mean gives always gives rise to lots of questions uh because of some of the interesting and uh at least initially counter-intuitive claims here but I don't want to spoil it so but as a kind of spoiler alert it kind gets to a really interesting but uh alas very brief discussion about the comparing these terms Precision attention and sensory attenuate and the relation and similarities and difference between these two these three terms and how each understanding each of them is essential to understanding the other ones but as I said it's a really interesting topic which gives rise to lots of discussions and I believe it's one of those topics that uh that's worth looking a bit more uh looking into in some other literature as well great well said what a cliffhanger next they go to a classic Model family called lockabletera these Dynamics inherit from characterizations of Predator prey Dynamics in ecology so it's kind of a classical ecology model shown in figure 8.2 on the top it's actually the ecosystem model plants herbivores and carnivores which follow different kinds of oscillatory Trends in continuous time and so that also has enabled it to be applied for other so-called winnerless competitions and that relates to topics like neural Darwinism and also neurodynamics where things have kind of oscillatory relationships with each other which are being modeled as a continuous time underlying process with a lot of measurement noise and discretization through space and time but those are the kinds of algorithms that SPM explores more and there's lack of Volterra and a lot of other dynamical systems theory in SPM so active inference kind of adds action and more to what was laid out from a pure dynamical systems theory in SPM here it really is just showing the ecology example and how you can project if you have three different species you can think about that motion in a cube or tetrahedron and then you could project onto kind of like looking at a lower dimensional manifold relating just two of the three species and that events is this kind of oscillatory but also moving Behavior that gets connected in figure 8.3 to neurobiology what would you say about uh okay so here in figure 8.3 uh we see some applications of um a lock of Altera Dynamics uh so uh the left column uh here uh show represents um what happens in I mean in eye blinking eye blink conditioning so uh of course here we need to account for uh I mean the expected States of uh of the of the sequel of the sequences of events that happens in in the eye blinking so uh the upper upper left figure shows the expectations uh in terms of time and then uh the parallel right hand side equation sorry right hand side figures uh shows the uh the lock of Altera system that that is applied um in the handwriting situation so as we can see although uh the uh I mean mathematical technology is the same or at least the modeling technology is the same the outcome of each situation uh varies drastically in two distinct the two distinct neurobiological Behavior not neurobiological but biological Behavior Uh so yeah we can see how the same modeling framework can give rise to different outcomes uh based on what parameters is needs to be optimized what parameters are selected for the modeling and so on so I believe it's quite uh interesting example to compare handwriting and the blinking together and how those can be compared to each other using the last couple Terror Dynamics great thank you box 8.2 gives a variance on the learning here presented with the formalism for continuous models kind of a technical aside section 8.4 is about generalized synchrony so figure 8.4 is going to visualize one of the classic dynamical systems which is the Lorenz attractor so what would you say about this bigger okay so this section is uh truly interesting because uh when one thinks of active inference probably the first situations that comes to mind is uh the situations in which we have a quite well-defined uh probability distributions for different parameters but as we can see here in section 8.4 actually some of the formalism of active inference can be successfully used to characterize even chaotic systems and partic and in particular the way in which two chaotic systems can be synchronized with each other so this is a classic example of a chaotic Lorenz system and it derives from it draws upon from some of Professor person's earlier work on Birdsong synchrony and as a side note any literature before 2016 is considered earlier history in active inference literature because it evolves uh quite rapidly so uh yeah this kind of synchrony between two chaotic systems uh can be interpreted as providing evidence or even let's say a way to model a kind of primitive theory of mind uh in in the sense that how exactly can we understand uh or can two two agents can trace each other's trajectories uh without um any I mean engaging in any direct exchange of observations between their internal and external States so uh yeah that's a really good example and I believe one of the most interesting examples of how active inference can even account for these kinds of uh Behavior so and the rest of the section goes into the details of how this kind of synchrony between um multi multi-scale uh Birds multi-scale Lorenz systems can happen um and how can we formulate it mathematically in terms of continuous time active inference awesome and there's been more recent work on Mark all blankets and stochastic chaos but the bird example is a classic 8.5 goes into hybrid discrete and continuous models so this could be kind of like a in-between chapter of seven and eight but now that we've been introduced to the pure form of discrete in the pure form of continuous models here shown the dot composability extends to so-called Hybrid models where here the lower level visually is using the continuous time formalism and the higher level is describing the little line added here the discrete time formalism and this was the similar structure described by the authors of the paper octave inference does not contradict folk psychology where they describe this lower level as motor active inference which was closely allied with the spinal Arc reflex shown above and then this higher level they call decision active inference because in that case it was referring to a discrete decision and so they used that kind of basic motif of continuous activity or continuous time modeling at the more peripheral aspects of a cognitive entity and like Ali said more discretization and hybridization as well at higher levels of the cognitive modeling and that type of an architecture here instead of describing who wants the ice cream cone I believe here it's gonna be a mixed or hybrid model that is going to call back the isocade system where there's a fixed point that is able to be moved as a set point and then there's a continuous time isocade that pursues the new fixed point and so that's analogous to a new set point or fixed point being specified from the top down muscle command about a new location for a muscle followed by movement towards it this is a muscular activity that is realizing that but but not in the elbow coming away from the hot stove this is about the eye cicading to an epistemic foraging location specified by top down hierarchical systems 8.3 describes little technical aside on mixture of gaussian gaussian mixture models kind of a technical modeling note an 8.6 closes it says it's a huge topic and much has been left out and so they list in table 8.1 key advances in continuous time models and those areas are synthetic bird song ocular motor delays conditioned reflexes smooth Pursuit eye movements psychosis illusions cicades action observation attention Hybrid models and self-organization and that's chapter 8. what else would you say and also what would you kind of lead someone to in the philosophical implications of eight because it sounds kind of cool uh well uh the case of continuous time active inference uh I think it leads to a really interesting questions uh both in terms of uh philosophical questions and also uh a more practical modeling questions about uh what parameters needs to be accounted for and so on uh and as I said I believe it's a more and more interesting way of um if not interesting but but at least more involved way of doing active inference modeling but uh one thing that uh one of the philosophical questions that when I uh have explored in our paper is how the processes of I mean ontological processes can uh philosophically described using fpp assertions in terms of their interaction with the environment [Applause] in which they co-constitute themselves and we don't necessarily distinguish between uh between the internal and the external state so one obvious example of this is that generalized synchrony example that we saw in this chapter in which we don't necessarily distinguish between which of the birds act as the agents and which one is the um environment or device versa so these kind of co-constitution of the environment and the agent which gives rise to the partitioning of State space through a markup blanket is uh one of the interesting uh philosophical points that I think needs to be elaborated a bit more using and some of the recent advances in philosophy such as the two the tools that's been developed in new materialism school or some other philosophical approach approaches but yeah these kinds of uh what exactly gives size gives rise to emergence what is the ontological status of emergent properties and so on are some of the burning questions for many philosophers today and I believe active inference in particularly continuous time active inference provides a clear uh precise mathematical formalism uh even if not to answer these questions but at least to explore it in a more rigorous and uh practical way and also practical and attractive attractable way so this is the area that I believe uh philosophy and science uh are beautifully intertwined uh into a coherent view of not only the phenomenon of interest but even about the whole world wow wow pretty cool yeah a lot to say about that topic after completing chapter seven and eight you've seen the kind of two major branches or two major motifs of just one kind of modeling but these kind of models have so many different forms that that's why it's such a Hands-On process to specify the generative model in chapter six and fit it with data in chapter nine those are all what's required and that's kind of the last mile of where these discussions about General motifs gets you but also playing with these pedagogical models can be really helpful because it will help you understand the basic patterns and relationships and start to see see different patterns in the graphical models and know from there what levels of technical processes can be kind of coarse grained over okay well that's it um I guess next time we will do probably 9 10 and maybe something else all right I'll end it now thanks Holly thank you